Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators

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1. Motivation

TPU pods succeed not because each chip is faster in isolation but because thousands of chips are wired together with a carefully designed memory and interconnect system. The most recent TPU7x/Ironwood pods publicly expose 9216 chips per pod, 192 GiB of HBM per chip, and 1200 GB/s of bidirectional inter-chip bandwidth per chip. For language model training and inference at this scale, the hard part is moving things—decode caches, expert tokens, gradients, layer-tied weights—rather than computing them.

Question. If a future TPU pod were wired with Lightmatter Passage-style photonic interconnect, would we just observe a lower-energy link, or would the optimal memory placement and distributed schedule above the link change as well?

2. Underlying Principles

Four conceptual layers. We separate the workload (an einsum), the architecture (per-chip memory plus a logical cluster-memory level), the inner mapping picked by the mapper, and the distributed schedule chosen above the mapper (point-to-point exchange, remote pooled-HBM streaming, replicated cache, photonic multicast cache fill).

Network model.

$$E_{\text{net}} = B\big(e_c + h\,\ell(N)\,e_{tx}\big), \quad T_{\text{net}} = \frac{B}{BW/C(N)} + h\,t_r + h\,\ell(N)\,t_p + t_s.$$

Electrical and chiplet links pay an energy term that grows with average path distance. Photonic links are conversion-dominated and stay nearly flat. Non-multicast baselines pay every receiver. Photonic multicast pays the unique payload exactly once.

Iterative loop. The mapper runs, its traffic is priced through the topology, and the resulting energy and latency are folded back into the logical-memory level. The mapper is then rerun until those per-bit numbers stabilize.

3. Evaluation Setup

- **Held fixed:** compute (MAC, register file, unified buffer) and per-chip HBM energy and latency.
- **Varied:** interconnect technology, topology, multicast support, and the distributed schedule.
- **Baselines:** an electrical TPU-style torus and a deliberately strong 3.5 Tb/s, 6 pJ/bit chiplet baseline so the comparison is not against straw electronics.
- **Photonics:** EVK100 (3.2 Tb/s class, 1.9–3.2 pJ/bit) and M1000 (57.3 Tb/s one-way, 2.3 pJ/bit including laser, multicast-capable optical circuit switching).
- **Workloads:** six representative GEMM proxies (decode MLP, KV attention, prefill projection, long-context prefill, mixture-of-experts dispatch, training weight gradient) plus public Gemma 4 anchors and an explicitly labeled long-context Gemini-class stress proxy.

4. Final Experiment Spine

Six experiments, one question each, in the order they were run

1. 16-node AccelForge feedback loop
Q: Does a logical-memory network level change energy and latency?
A: Yes: optical multicast reduces cluster energy by ~72% and latency by ~98%.

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2. Distance-aware link model
Q: Is pJ/bit constant with scale?
A: No: electrical/chiplet links grow with distance; photonics is conversion dominated.

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3. TPU cube mapping and multicast
Q: Is the gain just bandwidth?
A: No: shared-state multicast collapses N unicasts into one broadcast.

4. Mapping vs distributed schedule
Q: Can the AccelForge mapping stay stable while compiler choices change?
A: Yes: 0/24 inner signatures change, but 5/16 distributed choices change.

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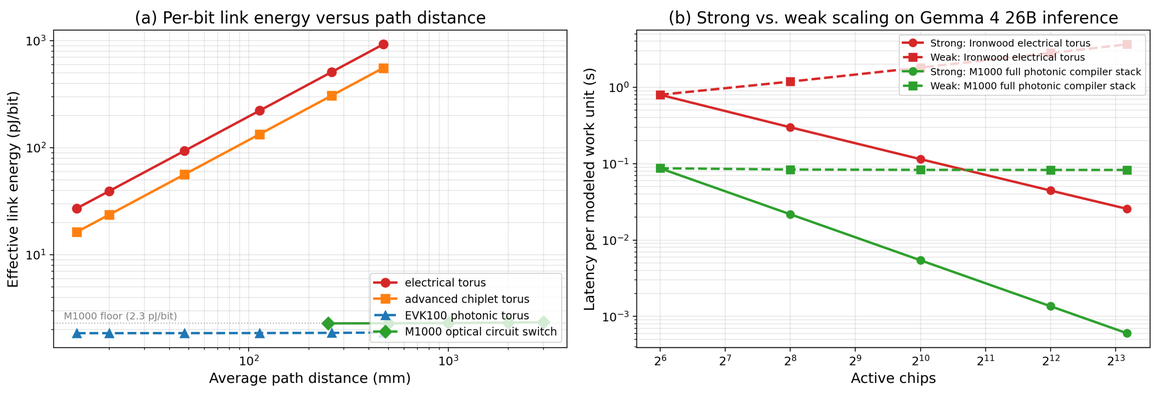
5. Strong/weak scaling and OCS/OPS
Q: What improves with larger systems and switching semantics?
A: Photonic multicast scales flatter; OPS mainly helps fine-grained phases.

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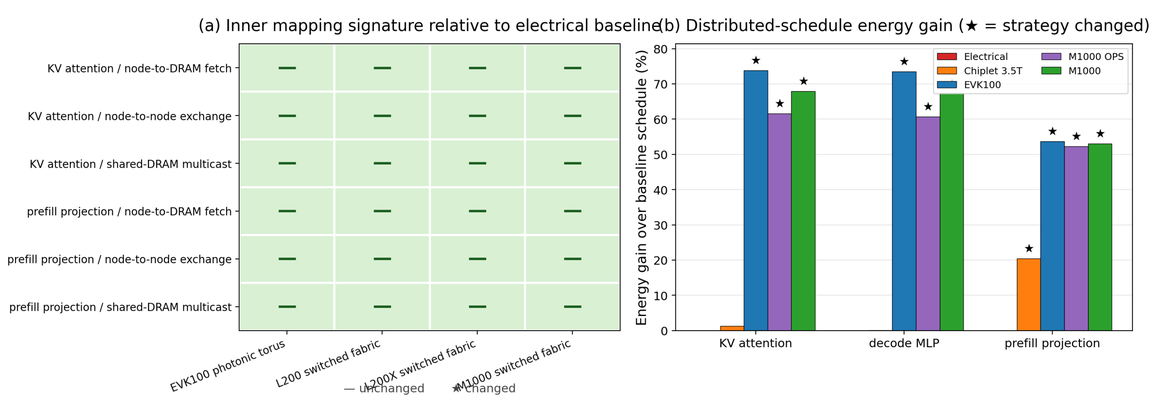
6. 9216-chip Gemma/Gemini proxy
Q: Does the stack matter end to end?
A: Inference improves far more than training because training becomes compute-bound.

5. Results: Distance Physics and Mapping

Distance physics and scaling behavior

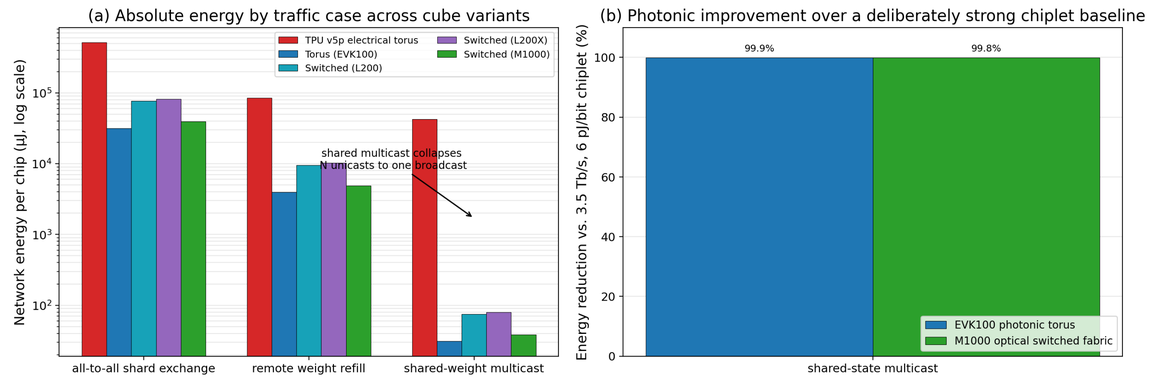


Local mapping is stable; the distributed schedule is what changes

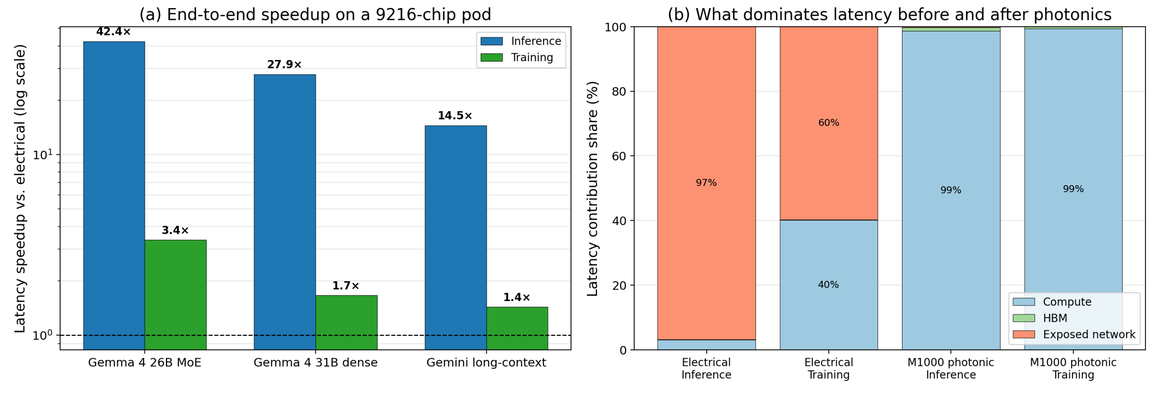


6. Results: Multicast and Pod Impact

Multicast and topology semantics, not raw bandwidth, are decisive

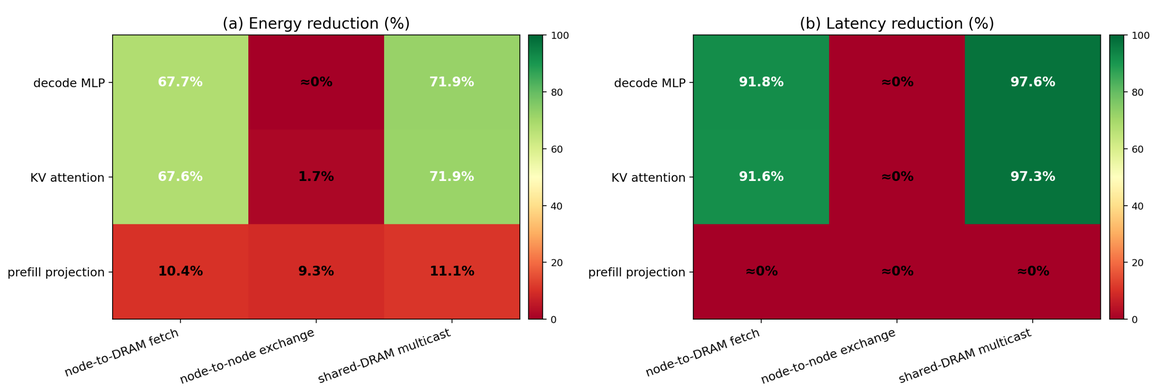


Pod-scale impact: inference benefits most; training shifts to a compute ceiling



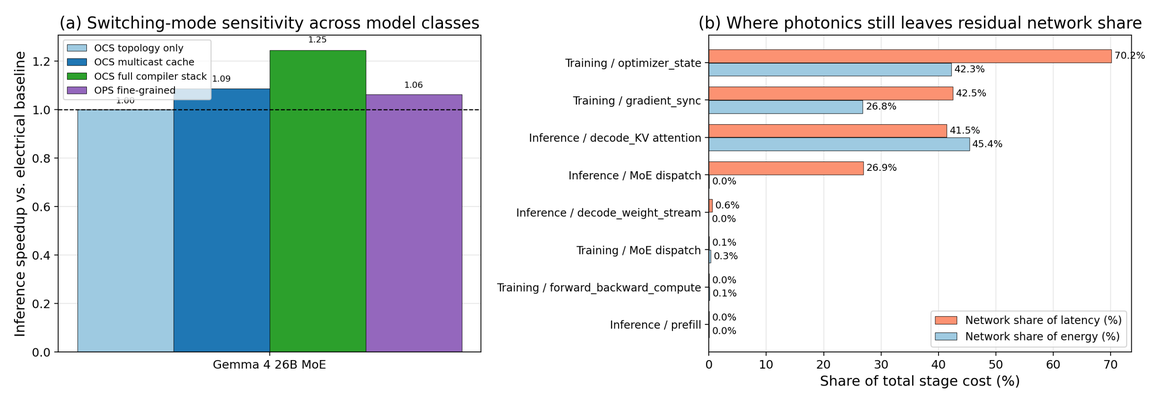
7. Cluster Validation Heatmap

Photonic crossbar over electrical mesh: 16-node cluster



8. Switching and Residual Bottlenecks

Switching mode and residual bottlenecks after photonics



9. Analysis of Results

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